

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Vector Functions and Parametric Curves II

Dr Francois Ebobisse Bille

0.1 The arclength reparameterization of a curve

Let $\mathcal{C}$ be a regular curve described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$. The arclength reparameterization of the regular curve $\mathcal{C}$ is a few steps process.

Define the arclength function $s: [a, b] \rightarrow [0, L(\mathcal{C})]$ as

$$s(t) := \int_a^t \|\vec{r}'(u)\| \, du \quad (1)$$

which is the length of the part of the curve $\mathcal{C}$ between $\vec{r}(a)$ and $\vec{r}(t)$.

Notice from the Fundamental Theorem of Calculus (FTC) that

$$\frac{ds}{dt} = \|\vec{r}'(t)\| > 0 \text{ for every } t \in (a, b). \quad (2)$$

Therefore, the function $s: [a, b] \rightarrow [0, L(\mathcal{C})]$ is one-to-one and hence, can be used in a change of parameterization of the curve as follows: $t \rightarrow s(t) \Leftrightarrow s \rightarrow t(s)$, i.e.,

$$\left\{ \begin{array}{ll} s: [a, b] & \rightarrow [0, L(\mathcal{C})] \\ t & \rightarrow s(t) \end{array} \right. \text{ is one-to-one with inverse } \left\{ \begin{array}{ll} t: [0, L(\mathcal{C})] & \rightarrow [a, b] \\ s & \rightarrow t(s) \end{array} \right. . \quad (3)$$

Using the derivative of the inverse function, we obtain that

$$\frac{dt}{ds}(s) = \frac{1}{\frac{ds}{dt}(t(s))} = \frac{1}{\|\vec{r}'(t(s))\|} \quad (\text{from (2)}). \quad (4)$$

Now, we get a new vector function $\vec{\gamma}: [0, L(\mathcal{C})] \rightarrow \mathbb{R}^n$ describing the curve $\mathcal{C}$ defined by

$$\vec{\gamma}(s) := \vec{r}(t(s)). \quad (5)$$

The new vector function $\vec{\gamma}$ is called the **arclength reparametrization of the curve $\mathcal{C}$** has the advantage that naturally arises from the shape of the curve and does not depend on a particular coordinates system.

For the Chain Rule (CR) we get differentiating $\vec{\gamma}$ that

$$\vec{\gamma}'(s) = \vec{r}'(t(s)) \frac{dt}{ds}(s) = \frac{\vec{r}'(t(s))}{\|\vec{r}'(t(s))\|} \Rightarrow \|\vec{\gamma}'(s)\| = 1. \quad (6)$$

Remark 0.1 It follows from the description above that the process of the arclength reparametrization of the regular $\mathcal{C}$ described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$ has the following steps:

- (1) Define the arclength parameter $s := \int_a^t \|\vec{r}'(u)\| du$.
- (ii) Find the inverse of the strictly increasing function $t \rightarrow s(t)$ and get $s \rightarrow t(s)$. Notice that it can be **hard** to find explicitly the inverse.
- (iii) Substitute t by $t(s)$ in the expression of the vector function $\vec{r}$ and get the new function $\vec{\gamma}$ of the parameter s .

Example 0.2 Find the arclength reparametrization of the $\mathcal{C}$ described by the vector function $\vec{r}: [-\pi, 2\pi/3] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := (-2t\sqrt{3}, \sin(2t), -\cos(2t)) \quad \forall t \in [-\pi, 2\pi/3].$$

Solution: We set $s := \int_{-\pi}^t \|\vec{r}'(u)\| du$.

Now,

$$\vec{r}'(t) = (-2\sqrt{3}, 2\cos(2t), 2\sin(2t)) \Rightarrow \|\vec{r}'(t)\| = 4 \Rightarrow s = 4(t + \pi) \Rightarrow t = \frac{s}{4} - \pi.$$

Now, as a final step we substitute t into $\vec{r}(t)$ and get

$$\vec{\gamma}(s) = \vec{r}\left(\frac{s}{4} - \pi\right) = \left(2\pi\sqrt{3} - \frac{s\sqrt{3}}{2}, \sin\left(\frac{s}{2} - 2\pi\right), -\cos\left(\frac{s}{2} - 2\pi\right)\right). \quad (7)$$

0.2 The curvature

Let $\mathcal{C}$ be a regular curve described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$. Intuitively, the **curvature** of the curve $\mathcal{C}$ at $\vec{r}(t)$ measures how “**curved**” or “**bent**” is the curve $\mathcal{C}$ at $\vec{r}(t)$. Precisely, the curvature measures how much the curve is bending as it is traced out or how much the curve is deviating from being a straight line at the given point. From this intuitive idea of a curvature, we can observe the description from the following picture.

The “bending” of the curve at a given point is measured through the variation of the direction of the tangent line with respect to the arclength. In other words, one can appreciate better the “bending” of the curve at a certain point, by moving along the arc of the curve around that point and observing the change in the direction of the tangent line w.r.t. that displacement. This is illustrated in figure 1 below.

As one moves along the curve $\mathcal{C}$ from A to B , the change in the direction of the tangent line is measured by $\Delta\theta$. In this case, the mean curvature of the arc AB of the curve $\mathcal{C}$ is given by

$$\left| \frac{\Delta\theta}{\Delta s} \right|.$$

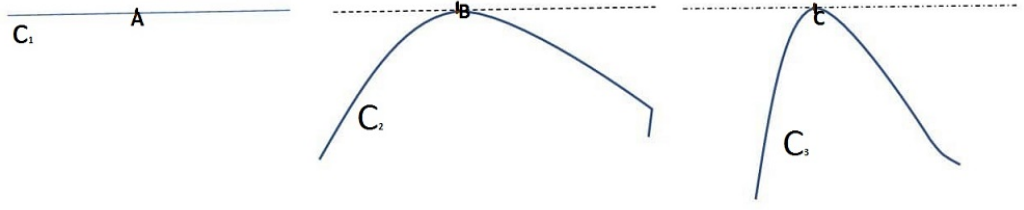


Figure 1: The curvature of C_1 at the point A as well as at any other point is equal to zero since the curve C_1 is just a straight line. The curvature of C_3 at the point C is higher than the curvature of C_2 at the point B , since the deviation of the curve C_3 at the point C from the horizontal line is higher than the one of the curve C_2 at the point B .

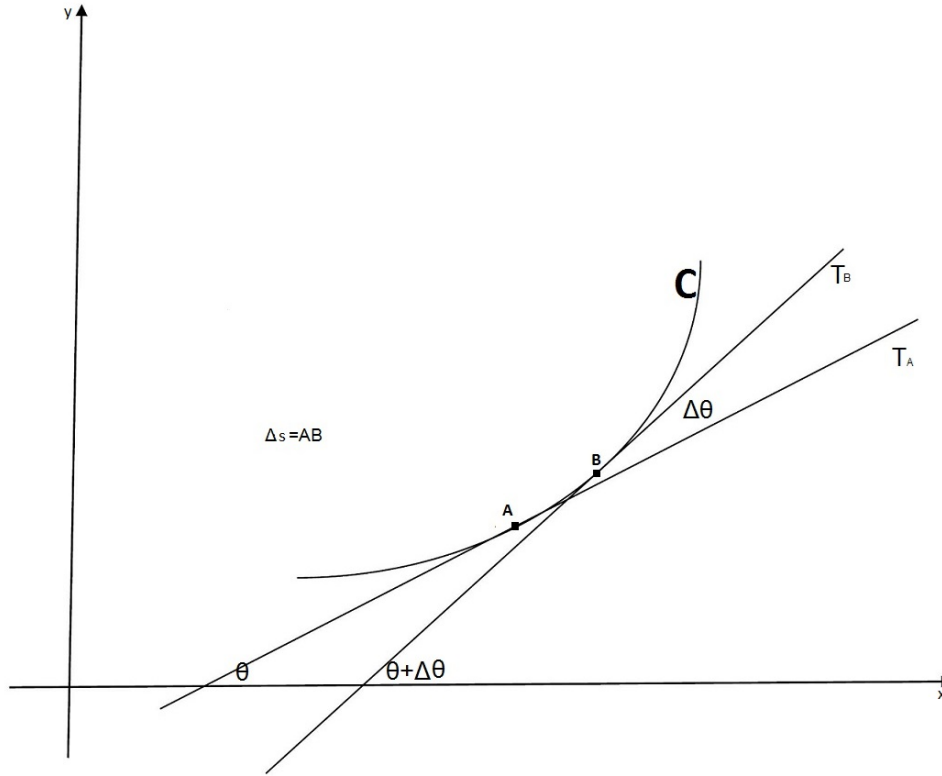


Figure 2: (1) Let Δs be the displacement along the curve from A to B . (2) Let T_A and T_B be respectively the tangent lines to the curve C and the points A and B respectively, which make the angles θ and $\theta + \Delta\theta$ with respect to the x -axis.

So, if the curve C is described by the arclength parametrization and we recall that $\vec{T}$ is the unit tangent vector, then the curvature is defined as

$$\kappa := \lim_{\Delta s \rightarrow 0} \left| \frac{\Delta\theta}{\Delta s} \right| = \left\| \frac{d\vec{T}}{ds} \right\|. \quad (8)$$

Whenever the regular curve $\mathcal{C}$ is described by a vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^3$, recalling the definition of the arclength parameter in Section 0.1, that is,

$$s = \int_a^t \|\vec{r}'(u)\| du \quad \Rightarrow \quad \frac{ds}{dt} = \|\vec{r}'(t)\| > 0,$$

we write

$$\frac{d\vec{T}}{ds} = \frac{\frac{d\vec{T}}{dt}}{\frac{ds}{dt}} = \frac{\vec{T}'(t)}{\|\vec{r}'(t)\|}.$$

Hence, we get that

$$\kappa(t) = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|}. \quad (9)$$

Example 0.3 Consider the curve $\mathcal{C}$ described by the vector function $\vec{r}: [-\pi, \pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := (\sin(2t), -3t, -\cos(2t)).$$

- (a) Find the unit tangent vector of the curve $\mathcal{C}$ at the point $\left(\frac{\sqrt{3}}{2}, -\frac{\pi}{2}, -\frac{1}{2}\right)$.
- (b) Find the curvature of the curve $\mathcal{C}$ at the point in (a).

Solution:

- (a) The point $\left(\frac{\sqrt{3}}{2}, -\frac{\pi}{2}, -\frac{1}{2}\right)$ corresponds to $t = \frac{\pi}{6}$. We want find $\vec{T}\left(\frac{\pi}{6}\right) = \frac{\vec{r}'\left(\frac{\pi}{6}\right)}{\|\vec{r}'\left(\frac{\pi}{6}\right)\|}$. Notice that

$$\vec{r}'(t) = (2\cos(2t), -3, 2\sin(2t)) \quad \Rightarrow \quad \vec{r}'\left(\frac{\pi}{6}\right) = (1, -3, \sqrt{3}) \quad \Rightarrow \quad \|\vec{r}'\left(\frac{\pi}{6}\right)\| = \sqrt{13}.$$

Thus,

$$\vec{T}\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{13}}(1, -3, \sqrt{3}).$$

- (b) We use the identity $\kappa\left(\frac{\pi}{6}\right) = \frac{\|\vec{T}'\left(\frac{\pi}{6}\right)\|}{\|\vec{r}'\left(\frac{\pi}{6}\right)\|}$. Now,

$$\begin{aligned} \vec{T}(t) &= \frac{1}{\sqrt{13}}(2\cos(2t), -3, 2\sin(2t)) \quad \Rightarrow \quad \vec{T}'(t) = \frac{1}{\sqrt{13}}(-4\sin(2t), 0, 4\cos(2t)) \\ \Rightarrow \quad \|\vec{T}'(t)\| &= \frac{4}{\sqrt{13}}. \end{aligned}$$

Thus,

$$\kappa\left(\frac{\pi}{6}\right) = \frac{\|\vec{T}'\left(\frac{\pi}{6}\right)\|}{\|\vec{r}'\left(\frac{\pi}{6}\right)\|} = \frac{4}{13}.$$

Remark 0.4 Sometimes $\|\vec{r}'(t)\|$ can be a complicated function of t and therefore the computation of $\vec{T}'(t)$ and hence, $\|\vec{T}'(t)\|$ can be tedious. For instance, let $\mathcal{C}$ be the curve described by the vector function $\vec{r}(t) := (\cos(2t), \cos t, \sin t)$ for every $t \in [0, 2\pi]$. Then,

$$\vec{r}'(t) = (-2\sin(2t), -\sin t, \cos t) \quad \text{and} \quad \|\vec{r}'(t)\| = \sqrt{4\sin^2(2t) + \sin^2 t + \cos^2 t} = \sqrt{4\sin^2(2t) + 1}.$$

Hence,

$$\vec{T}(t) = \left(\frac{-2\sin(2t)}{\sqrt{4\sin^2(2t) + 1}}, \frac{-\sin t}{\sqrt{4\sin^2(2t) + 1}}, \frac{\cos t}{\sqrt{4\sin^2(2t) + 1}} \right).$$

As you can see, computing the derivative of $\vec{T}$ and hence its magnitude as required in the first formula of the curvature, is **unpleasant!**

When you are in such a situation, it is recommended to use an alternative formula for the computation of the curvature that will be established in the next theorem.

Theorem 0.5 Let $\mathcal{C}$ be a regular curve described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^3$. Assume that the function $\vec{r}$ is twice continuously differentiable. Then the curvature of the curve $\mathcal{C}$ at $\vec{r}(t)$ is

$$\kappa(t) = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}. \quad (10)$$

We omit the proof of this theorem in these notes!

Remark 0.6 We have two formulas for the curvature. However, it is important to know which one is more convenient to use under which conditions.

- If $\|\vec{r}'(t)\|$ is a constant function of t , then you can use the formula $\kappa(t) = \frac{\|\vec{T}'(t)\|}{\|\vec{r}'(t)\|}$.
- If $\|\vec{r}'(t)\|$ is not a constant function of t , it is more convenient to use the formula with the cross-product, that is, $\kappa(t) = \frac{\|\vec{r}'(t) \times \vec{r}''(t)\|}{\|\vec{r}'(t)\|^3}$.

Exercises 0.7

1. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: [-\frac{\pi}{2}, \frac{5\pi}{4}] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := (t + t \cos t - \sin t, \cos(2t), \cos t + t \sin t) \quad \forall t \in [-\pi/2, 5\pi/4].$$

- (a) Show that the curve $\mathcal{C}$ is regular.
- (b) Find the unit tangent vector at the point $(0, 1, -1)$.
- (c) Show that the curvature of the curve $\mathcal{C}$ at the point in (b) is equal $\frac{\sqrt{\pi^4 + 14\pi^2 + 17}}{(1 + \pi^2)^{3/2}}$.
- (d) Write down the cartesian equation of the normal plane to the curve $\mathcal{C}$ at the point in (b).

2. Let $\mathcal{C}$ be the curve described by the vector function $\vec{r}: \mathbb{R} \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := (e^t \cos t, e^t \sin t, 2) \quad \forall t \in \mathbb{R}.$$

Show that the curvature of the curve $\mathcal{C}$ at every point $\vec{r}(t)$ is $\kappa(t) = \frac{1}{\sqrt{2}}e^{-t}$.

Remark 0.8 (The curvature of a graph)

Let $f: [a, b] \rightarrow \mathbb{R}$ be a continuously twice differentiable function. We want to find the curvature $\kappa(x)$ of the curve $\mathcal{C}$ graph of the function f at the point $(x, f(x))$. The curve $\mathcal{C}$ is described by the vector function $\vec{r}(x) = (x, f(x), 0)$ when viewed as a curve in the space $\mathbb{R}^3$. So, we find

$$\vec{r}'(x) = (1, f'(x), 0) \quad \Rightarrow \quad \vec{r}''(x) = (0, f''(x), 0).$$

Hence,

$$\vec{r}'(x) \times \vec{r}''(x) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & f'(x) & 0 \\ 0 & f''(x) & 0 \end{vmatrix} = (0, 0, f''(x))$$

Thus,

$$\kappa(x) = \frac{\|\vec{r}'(x) \times \vec{r}''(x)\|}{\|\vec{r}'(x)\|^3} = \frac{|f''(x)|}{(1 + f'(x)^2)^{3/2}}. \quad (11)$$

0.3 The normal and binormal vectors

Let $\mathcal{C}$ be a regular curve described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^3$. We defined the unit tangent vector as

$$\vec{T}(t) := \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}.$$

Also,

$$\begin{aligned} \|\vec{T}(t)\| = 1 &\Leftrightarrow \|\vec{T}(t)\|^2 = 1 \Leftrightarrow \vec{T}(t) \cdot \vec{T}(t) = 1 && \xRightarrow{\text{Differentiating}} 2\vec{T}(t) \cdot \vec{T}'(t) = 0 \\ &&& \Leftrightarrow \boxed{\vec{T}(t) \perp \vec{T}'(t)}. \end{aligned}$$

Notice that there are infinitely many vectors which are orthogonal to the vector $\vec{T}(t)$. **However, we want to select a special vector with the geometrical property that, for a non-straight line, it points in the turning direction of the curve at the given point.**

Definition 0.9 (The principal unit normal vector)

Let $\mathcal{C}$ be a regular curve described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^3$. The **principal unit normal vector** to the curve $\mathcal{C}$ at $\vec{r}(t)$ is defined as

$$\vec{N}(t) := \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|} = \frac{1}{\kappa(t)} \frac{\vec{T}'(t)}{\|\vec{r}'(t)\|}. \quad (12)$$

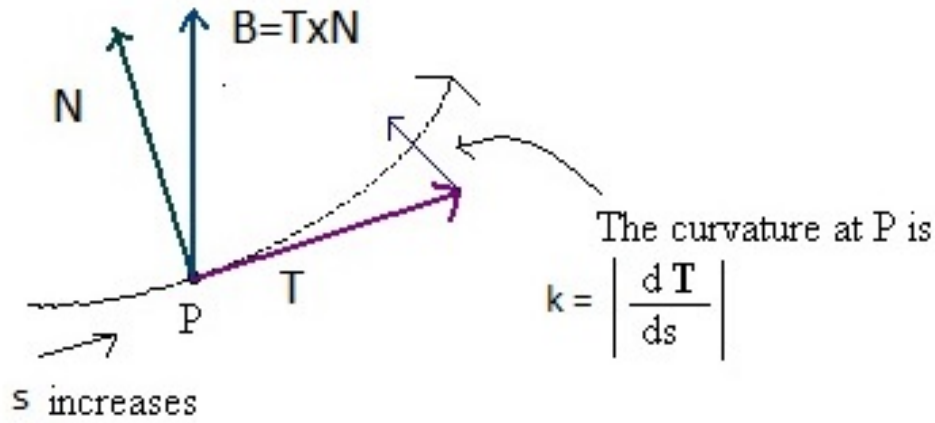


Figure 3:

Definition 0.10 (The binormal vector)

Let $\mathcal{C}$ be a regular curve described by the vector function $\vec{r}: [a, b] \rightarrow \mathbb{R}^3$. The binormal vector to the curve $\mathcal{C}$ at $\vec{r}(t)$ is defined as

$$\vec{B}(t) := \vec{T}(t) \times \vec{N}(t). \quad (13)$$

Definition 0.11

- (i) The plane spanned by the vectors $\vec{T}(t)$ and $\vec{N}(t)$ is called the **osculating plane** to the curve $\mathcal{C}$ at $\vec{r}(t)$. The normal vector to the osculating plane is $\vec{B}(t)$.
- (ii) The plane spanned by the vectors $\vec{N}(t)$ and $\vec{B}(t)$ is called the **normal plane** to the curve $\mathcal{C}$ at $\vec{r}(t)$. The normal vector to the normal plane is $\vec{T}(t)$.
- (iii) The plane spanned by the vectors $\vec{B}(t)$ and $\vec{T}(t)$ is called the **rectifying plane** to the curve $\mathcal{C}$ at $\vec{r}(t)$. The normal vector to the rectifying plane is $\vec{N}(t)$.
- (iv) The triple $(\vec{T}(t), \vec{N}(t), \vec{B}(t))$ is called the **Frenet-Serret frame** of the curve $\mathcal{C}$ at $\vec{r}(t)$.

Example 0.12 In the example 0.3, we considered the curve $\mathcal{C}$ described by the vector function $\vec{r}: [-\pi, \pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(t) := (\sin(2t), -3t, -\cos(2t)) \quad \forall t \in [-\pi, \pi].$$

- (a) Find the curvature of the curve $\mathcal{C}$ at the point $(0, -\frac{3\pi}{2}, 1)$.
- (b) Find the Frenet-Serret frame $(\vec{T}, \vec{N}, \vec{B})$ of the curve $\mathcal{C}$ at the point in (a).
- (c) Write down the cartesian equation of normal plane of $\mathcal{C}$ at the point in (a).

Solutions:

(a) The point $(0, -\frac{3\pi}{2}, 1)$ corresponds to $t = \frac{\pi}{2}$. So, we need

$$\vec{r}'(t) = (2 \cos(2t), -3, 2 \sin(2t)) \Rightarrow \|\vec{r}'(t)\| = \sqrt{13} \Rightarrow \boxed{\vec{T}(t) = \frac{1}{\sqrt{13}}(2 \cos(2t), -3, 2 \sin(2t))}.$$

So,

$$\vec{T}'(t) = \frac{1}{\sqrt{13}}(-4 \sin(2t), 0, 4 \cos(2t)) \Rightarrow \|\vec{T}'(t)\| = \frac{4}{\sqrt{13}}.$$

Hence,

$$\kappa\left(\frac{\pi}{2}\right) = \frac{\|\vec{T}'(\frac{\pi}{2})\|}{\|\vec{r}'(\frac{\pi}{2})\|} = \frac{4}{13}.$$

(b)

$$\boxed{\vec{T}\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{13}}(-2, -3, 0), \quad \vec{N}\left(\frac{\pi}{2}\right) = \frac{\vec{T}'(\frac{\pi}{2})}{\|\vec{T}'(\frac{\pi}{2})\|} = (0, 0, -1)}$$

and

$$\boxed{\vec{B}\left(\frac{\pi}{2}\right) = \vec{T}\left(\frac{\pi}{2}\right) \times \vec{N}\left(\frac{\pi}{2}\right) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2/\sqrt{13} & -3/\sqrt{13} & 0 \\ 0 & 0 & -1 \end{vmatrix} = (3/\sqrt{13}, -2/\sqrt{13}, 0)}$$

(c) By definition the normal plane of $\mathcal{C}$ at the point in (a) has normal vector $\vec{T}(\frac{\pi}{2})$ or $\vec{r}'(\frac{\pi}{2})$. Hence, the normal plane has cartesian equation

$$\boxed{(x - 0, y + \frac{3\pi}{2}, z - 1) \cdot (-2, -3, 0) = 0 \Leftrightarrow -2x - 3(y + \frac{3\pi}{2}) + 0(z - 1) = 0}.$$

0.4 Motion of objects in the space

As an application of the concepts which arose in the study of curves such as the tangent, normal vectors, the arclength, the curvature, we study in this section the motion of objects on a plane and in the space.

First of all when an object is moving, we use a vector function to represent the position of the object as a function of the time. Let us notice that in the definition of a curve described by a vector function, one expects that vector function to be at least continuous. This is motivated somehow motivated by the fact that, a moving object cannot make “jumps in time” i.e., it arrives at some point at a certain instant, it can only leave from that point!!!

Let $\vec{r}(t)$ be the position of the object as a function of time. We obtain then a vector function $\vec{r}$. Unlike the position vector function $\vec{r}$, the velocity vector function can be discontinuous. This is the case when for instance “**shocks**” happen during the motion of the object.

Assume that the position function $\vec{r}$ is twice differentiable. Then the velocity of the object at the instant t is the rate of the position function at that instant i.e.,

$$\vec{v}(t) := \vec{r}'(t),$$

the acceleration vector of the object at the instant t is the rate of change of the velocity at that instant i.e.,

$$\vec{a}(t) := \vec{v}'(t) = \vec{r}''(t)$$

and the speed of the object at the instant t is the magnitude of the velocity at that instant i.e.,

$$c(t) := \|\vec{v}(t)\|.$$

Example 0.13 Find the velocity, acceleration and speed of a particle with position vector

$$\vec{r}(t) := (t^2, 3t - 2, e^{t+1})$$

Solution:

$$\begin{aligned}\vec{v}(t) &= \vec{r}'(t) = (2t, 3, e^{t+1}) \\ \vec{a}(t) &= \vec{v}'(t) = (2, 0, e^{t+1}) \\ c(t) &= \|\vec{v}(t)\| = \sqrt{4t^2 + 9 + e^{2(t+1)}}.\end{aligned}$$

Exercise 0.14 (Motion of a fallen object) A ball is thrown horizontally at a speed of 90 m/s from the top of a building with height 25 meters from the ground. Assume that air resistance is negligible while the ball is travelling in the air.

- (i) Find the position $\vec{r}(t)$ of the ball at every instant t .
- (ii) How much time does the ball spend in the air?
- (iii) Find the total distance travelled by the ball as a definite integral of some function.
- (iv) Where does the ball hit the ground?
- (v) What is the speed of the ball at the time it hits the ground?

Solution: The initial position of the ball is $\vec{r}(0) = (0, 25)$ while the initial velocity is $\vec{v}(0) = (90, 0)$. If the air resistance is negligible, the ball is moving in the air with the only attached force $\vec{P} = -mg\vec{j}$ where m is the mass of the ball and $g = 9.8 \text{ m.s}^{-2}$ is the acceleration due to gravity.

- (i) According to Newton's second law of motion, the resultant of the attached forces equals the inertia. So, we get the equation

$$\underbrace{-mg\vec{j}}_{\text{The only attached force}} = \underbrace{m\vec{a}(t)}_{\text{The inertia}} \Rightarrow \vec{a}(t) = -g\vec{j}.$$

Now

$$\vec{a}(t) = -g\vec{j} \Rightarrow \vec{v}(t) = \int_0^t \vec{a}(s)ds + \vec{v}(0) \Rightarrow \vec{v}(t) = -g\vec{j}t + \vec{v}(0).$$

Therefore we find that the position of the ball at every instant t is

$$\vec{r}(t) = \int_0^t \vec{v}(s) ds + \vec{r}(0) \quad \Rightarrow \quad \vec{r}(t) = -\frac{1}{2}gt^2\vec{j} + \vec{v}(0)t + \vec{r}(0) .$$

In components

$$\begin{cases} x = 90t, \\ y = -\frac{1}{2}gt^2 + 25 \end{cases} .$$

(ii) We seek t such that $y(t) = 0$. That is,

$$-\frac{1}{2}gt^2 + 25 = 0 \quad \Leftrightarrow \quad t = \sqrt{\frac{50}{g}} = \sqrt{\frac{50}{9.8}} \approx 2.26 \text{ s} .$$

(iii) The total distance travelled by the ball is the length of the curve traced by the object during its motion. That is,

$$D = \int_0^{2.26} \|\vec{r}'(t)\| dt = \int_0^{2.26} \|\vec{v}(t)\| dt = \int_0^{2.26} \sqrt{90^2 + 9.8^2 t^2} dt \quad \text{since } \vec{v}(t) = (90, -9.8t) .$$

(iv) The ball hits the ground at $\vec{r}(2.26) = (90 \times 2.26, 0) = (203.4, 0)$.

(v) The speed of the ball at the time it hits the ground is

$$c(2.26) = \|\vec{v}(2.26)\| = \sqrt{9.8^2(2.26)^2 + 90^2} \approx 92.69 \text{ m.s}^{-1} .$$

We will come back later on the parametric curves in the last chapter on Vector Calculus with the concept of line integrals!!!

In the next set of notes we will start the study of functions of several variables with domain, graph, limits and continuity!!!